Parasitology in Microbiology: Beginner's Course

Lesson 6: Host-Parasite Interactions

Host Parasite Interactions – Study Notes

Key Concepts

1. **Innate and adaptive immune responses to parasites** – how the host detects and fights infection.

2. **Pathogenic mechanisms employed by parasites** – ways parasites obtain nutrients, evade immunity, and cause damage.

3. **Link between parasite activity and symptom development** – why clinical signs appear and how they relate to host parasite dynamics.

Summary

Parasites are organisms that live on or inside a host and obtain nutrients at the host's expense. The host's immune system mounts a two tiered defense: the **innate response** (physical barriers, phagocytes, complement, inflammation) provides rapid, non specific protection, while the **adaptive response** (B cell antibodies, T cell mediated immunity) offers a slower but highly specific attack that can generate immunological memory.

To survive, parasites have evolved a suite of **pathogenic mechanisms**. These include:

• **Immune evasion** (antigenic variation, secretion of immunomodulatory molecules, mimicry of host proteins).

• **Nutrient acquisition** (e.g., sucking blood, degrading host tissues, hijacking host metabolic pathways).

• **Physical damage** (attachment structures that injure epithelia, migration through tissues).

The interaction of these mechanisms with host defenses produces the **clinical symptoms** we observe. For instance, inflammation caused by immune activation can lead to fever, pain, and swelling, while direct tissue destruction may cause anemia, malabsorption, or organ dysfunction. Understanding the balance between host resistance and parasite virulence is essential for diagnosing, treating, and preventing parasitic diseases.

Important Points

• **Innate immunity**: skin/mucosa, mucus, gastric acid, phagocytes, natural killer cells, complement cascade. Acts within minutes–hours.

• **Adaptive immunity**:

- *Humoral*: IgM !' IgG, IgE (important for helminths).
- *Cell mediated*: Th1 (intracellular parasites), Th2 (extracellular helminths).

• **Parasite evasion tactics**:

- Antigenic shift/variation (e.g., *Plasmodium falciparum* PfEMP1).
- Secretion of cytokine like proteins (e.g., *Schistosoma* IL 10 homolog).

- Surface coat shedding (e.g., *Trypanosoma* VSG coat).

• **Pathogenic mechanisms**:

- **Mechanical injury** – attachment organs (hooks, suckers) damage tissues.
- **Toxic metabolites** – excretory secretory products that disrupt host cells.
- **Nutrient theft** – blood feeding (mosquitoes, leeches), intestinal absorption (tapeworms).

• **Symptom development** often reflects host response more than direct parasite damage (e.g., fever from cytokine release).

• **Immunopathology**: excessive or misdirected immune responses can cause more harm than the parasite itself (e.g., cerebral malaria, eosinophilic meningitis).

Examples

Parasite	Host Interaction	Mechanism & Resulting Symptoms
Plasmodium spp. (malaria)	Invades red blood cells; triggers inflammatory cytokines	*Cytoadherence* of infected RBCs to endothelium !’ microvascular blockage; immune mediated fever, chills, anemia, splenomegaly.
Schistosoma mansoni	Adult worms reside in mesenteric veins; eggs embolize to liver/intestine	Eggs release proteases !’ granuloma formation; chronic abdominal pain, diarrhea, hepatosplenomegaly, portal hypertension.

Quick Review

1. **What are the two main arms of the host immune system that combat parasites, and how do they differ in timing and specificity?**
2. **Name two strategies parasites use to evade the host immune response.**
3. **Why can symptoms such as fever be considered a product of the host rather than the parasite itself?**
4. **Give one example of a mechanical pathogenic mechanism and the disease it causes.**
5. **How does a Th2 dominant response help the host fight helminth infections?**

Further Reading

• **Immunology of Parasitic Infections** – detailed pathways of Th1 vs. Th2 responses.

• **Antigenic Variation in Protozoa** – molecular mechanisms behind immune evasion.

• **Host Parasite Coevolution** – evolutionary arms race and its impact on disease epidemiology.

• **Immunopathology in Parasitic Diseases** – cases where the immune response causes tissue damage.

Use these notes as a foundation; supplement with textbook chapters or primary literature for deeper understanding.